

Towards Increased Detection Range of Magnetic Nanoparticles Using a Surface Acoustic Wave Magnetoelastic Sensor

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Towards Bio Sample Measurements

- Magnetic nanoparticles (MNPs) are key tools for magnetic resonance imaging contrast enhancement, targeted drug delivery and magnetic hyperthermia [1].
- Non-invasive detection and quantification of MNPs at biologically relevant distances remain challenging [2].
- Our group uses a magnetoelastic surface acoustic wave (SAW) sensor to quantify MNPs, but the detection range is limited when using conventional single-coil excitation.
- In this work, we test whether a Helmholtz coil configuration extends the usable detection distance and enables measurements in bio samples.

Sample holder with
Magnetic Nano particles

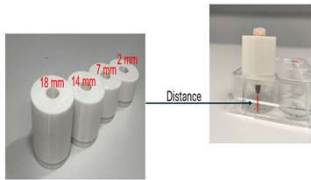
Six Well Plate

Surface Acoustic
Magnetoelastic
Sensor

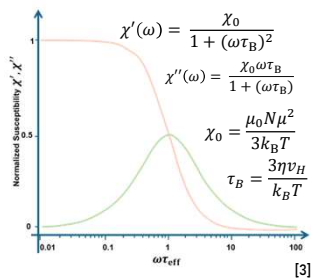
Helmholtz
coil pair

AC Susceptibility Setup

Distance Measurements

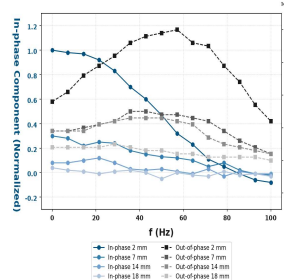


Measurement Principle



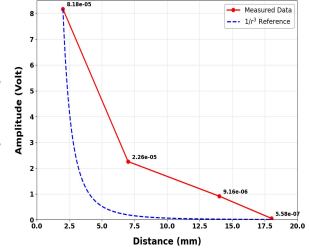
Distance Measurement Data

Combined In-phase and Out-of-phase Components



Amplitude vs Distance

Amplitude vs Distance at 163.6 Hz



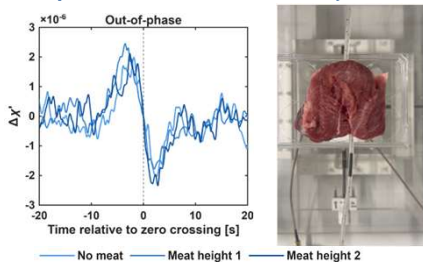
- Changing the sample holder size to increase the distance from size 2 mm to 18 mm.

η : Dynamic viscosity
 χ : Magnetic susceptibility
 k_B : Boltzmann constant
 N : Particle number density
 μ_0 : Magnetic field constant
 μ : Magnetic moment
 T : Temperature
 τ_B : Brownian relaxation time
 V_H : Hydrodynamic volume

- Signal amplitude decrease with increasing sensor-to-sample distance.

Bio sample Measurements

Pump Measurements with turkey breast meat

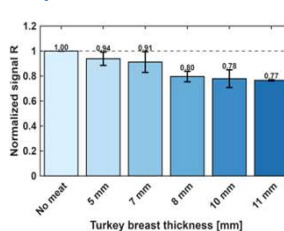


- Presence of turkey breast meat doesn't significantly influence MNPs signals.

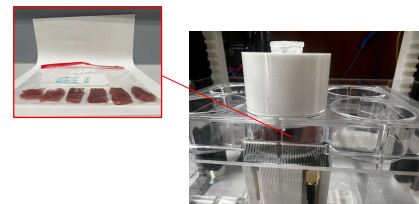
References

- [1] Krishnan, IEEE Trans. Magn., 46, 2523–2558 (2010).
[2] Barbieri et al., IEEE Sens. J., 26, 3812–3822 (2026).
[3] Ludwig, F., & Remmer, H. Physical Sciences Reviews, 7(9), 981–1008 (2022).

Amplitude vs Meat thickness



Different thickness of meat samples



- Increasing thickness has direct impact on signal amplitude.
- Further investigation is needed.

Conclusions & Outlook

- MNPs signal decreases with distance and tissue thickness.
- MNPs remain detectable through tissue.
- » Explore conductive layers for local AC-field tailoring and improved MNPs detection.

